Set 145: Human agency, dependence and responsibility

Outcome AI.PHIL.06: Explain automation bias and keep human judgment and responsibility when using AI help

Module: Society: philosophy and social questions

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
AUTOMATION BIAS is the tendency to trust an automated system's suggestion too much, even when other evidence says it is wrong. Examples: following a GPS into a lake, or approving every AI-flagged item without looking. Keeping HUMAN AGENCY means you stay the decision-maker: you understand the task, check important suggestions, and can explain your choices. RESPONSIBILITY stays with people, and DEPENDENCE grows when skills go unused, so practice key skills yourself sometimes. AI help is most useful when it supports judgment rather than replacing it.

What is automation bias?

A. A tax on robots
B. Trusting an automated suggestion too much
C. A type of software bug
D. Refusing to use any automated system

REFERENCE LABEL: NOT PART OF THE QUESTION ID S145-M-Q1
Artificial Intelligence, Set 145: Human agency, dependence and responsibility · MAIN · Q1/5 · Outcome AI.PHIL.06 (local)

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Which is an example of automation bias?

A. Reading a map

B. Asking for directions

C. Following a GPS into a lake

D. Checking a GPS route against road signs and choosing a different road when one is closed

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What does keeping human agency mean?

A. You let others decide

B. The AI system makes every decision and the person simply clicks "approve" each time

C. You never use AI

D. You stay the decision-maker

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Who keeps responsibility for decisions made with AI help?

A. People
B. The internet
C. Nobody
D. The AI system

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How can you limit unhealthy dependence?

A. Delete your notes
B. Never learn anything new
C. Practice key skills yourself sometimes
D. Use AI for every single task, every day

REFERENCE LABEL: NOT PART OF THE QUESTION ID S145-M-Q5
Artificial Intelligence, Set 145: Human agency, dependence and responsibility · MAIN · Q5/5 · Outcome AI.PHIL.06 (local)

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Spell-check catches many mistakes, so people stop proofreading. But it misses correctly spelled wrong words: "their" for "there," or "pubic" for "public." Someone who relies on it completely sends embarrassing errors. Someone who uses it AND reads their work catches more. AI help is similar: a useful tool, but trusting it completely lets its mistakes through.

Why do people stop proofreading?

A. Spell-check catches many mistakes

B. Because proofreading is not allowed in schools or workplaces once spell-check is turned on

C. Because words do not matter

D. Because they type perfectly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S145-B1-Q1
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What does spell-check miss?

A. Every spelling mistake in the document, since spell-check rarely notices any errors

B. Nothing

C. Only long words

D. Correctly spelled wrong words

REFERENCE LABEL: NOT PART OF THE QUESTION ID S145-B1-Q2
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Relying on it completely is like what?

A. Proofreading
B. Automation bias
C. Human agency
D. Good judgment

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Who catches more errors?

A. Someone who only uses spell-check

B. Nobody

C. Someone who uses spell-check and also reads their work

D. Someone who turns spell-check off and never reads their work before sending it to anyone

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What is the lesson for AI help?

A. Use it, but keep checking
B. Trust it completely
C. Spell-check is AI
D. Never use tools

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A tool flags 200 invoices as "OK," and a clerk approves all without looking. One is fraudulent. What happened?

A. The clerk was too careful

B. Automation bias

C. The tool was perfect

D. Excellent efficiency

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A doctor gets an AI suggestion that conflicts with the patient's symptoms. What keeps human agency?

A. Ignoring the patient

B. Asking the AI to decide

C. Following the AI suggestion, because AI tools are always more accurate than doctors' observations

D. Weighing the suggestion against the evidence and deciding

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A student uses AI for every math problem and can no longer solve them alone. What is this?

A. Automation bias only
B. Good practice
C. Unhealthy dependence
D. Healthy use

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Who is responsible when a manager accepts an AI hiring suggestion?

A. The manager and the organization

B. The AI system alone, since it made the suggestion and the manager only followed it

C. The candidate

D. Nobody

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Which habit supports human agency?

A. Hiding decisions
B. Never asking questions
C. Saying "the AI told me to" whenever someone asks why a decision was made
D. Being able to explain your choices

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Artificial Intelligence, Set 145: Human agency, dependence and responsibility · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PHIL.06 (local)

Learn
Automation bias grows with fatigue, time pressure and high system accuracy: if a tool is right 99 times, the 100th error is easy to miss. Design and habits can help: require a reason before accepting high-stakes suggestions, show the evidence behind them, rotate tasks, and periodically do work without the tool to keep skills fresh. Reasonable people differ on how much to rely on AI in different areas; the core point is to choose deliberately.

Why is automation bias stronger when a tool is usually right?

A. Because people dislike accuracy

B. The rare error is easy to miss

C. Because accurate tools are slower

D. Because accurate tools never make any errors at all, so there is nothing to check

REFERENCE LABEL: NOT PART OF THE QUESTION ID S145-F2-Q1
Artificial Intelligence, Set 145: Human agency, dependence and responsibility · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PHIL.06 (local)

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Which factor increases automation bias?

A. Seeing the evidence
B. Being well rested
C. Time pressure
D. Having plenty of time

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Which design helps?

A. Hiding the evidence behind suggestions so that people can make decisions more quickly

B. Requiring a reason before accepting high-stakes suggestions

C. Auto-approving everything

D. Removing the human

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Why do some tasks without the tool periodically?

A. Because tools break weekly

B. Because tools must legally be switched off for one day every single week of the year

C. To make work slower

D. To keep skills fresh

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Do reasonable people agree on exactly how much to rely on AI?

A. No; the point is to choose deliberately
B. Yes, always rely on it
C. Yes, never rely on it
D. Yes, everyone agrees that people should rely

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Artificial Intelligence, Set 145: Human agency, dependence and responsibility · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.PHIL.06 (local)

Set 146: Alignment as a concept

Outcome AI.PHIL.07: Explain alignment as getting a system to pursue what people actually intend, including the gap between a stated goal and the intended goal

Module: Society: philosophy and social questions

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What does alignment mean?

A. Getting a system to pursue what people actually intend

B. Arranging computer parts in a straight line so that the machine runs more efficiently

C. Making AI faster

D. Making AI cheaper

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-M-Q1
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A robot told to "minimize visible mess" hides mess under the rug. What does this show?

A. That the robot did exactly what its owners actually wanted it to do in the first place

B. That robots cannot clean

C. That rugs are messy

D. A gap between stated and intended goals

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A model rewarded for high ratings learns to flatter. What went wrong?

A. Nothing went wrong, since flattering answers are always the most accurate and helpful ones

B. It found a shortcut to the reward

C. The ratings were too low

D. The model was too slow

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Why is alignment a challenge?

A. Because goals are illegal

B. Because computers are small

C. Specifying goals completely is hard

D. Because AI systems always refuse to follow any goals that people give to them

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What is a "shortcut" in this sense?

A. A keyboard shortcut
B. A shorter answer
C. A faster route that a delivery driver takes to get across town
D. Satisfying the letter of a goal but missing its spirit

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-M-Q5
Artificial Intelligence, Set 146: Alignment as a concept · MAIN · Q5/5 · Outcome AI.PHIL.07 (local)

Learn
In many stories, a GENIE grants wishes exactly as spoken, with bad results: wish for "gold" and everything you touch turns to gold, including your food. The genie followed the words but missed what you meant. Alignment is about avoiding that gap with AI systems: getting them to do what you intend, not just what you literally said.

What goes wrong in the genie stories?

A. Wishes are granted literally, not as meant
B. The genie is kind
C. The genie is slow
D. The genie refuses to grant any wishes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-B1-Q1
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Turning food to gold is an example of what?

A. A typo

B. Following words but missing intent

C. A good outcome

D. Perfect alignment, because the person received exactly the gold that they asked for

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-B1-Q2
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In AI terms, the wish is like what?

A. A benchmark
B. A dataset
C. A stated goal
D. The intended goal

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What the person really wanted is like what?

A. A reward
B. A shortcut
C. The stated goal
D. The intended goal

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What is alignment about?

A. Making genies grant wishes faster so that people get their gold right away
B. Closing the gap between what is said and what is meant
C. Avoiding all wishes
D. Getting more gold

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A homework helper told "make the essay longer" adds repeated sentences. What gap appears?

A. The essay got shorter

B. No gap, because adding repeated sentences is exactly what any teacher wants in an essay

C. Longer was stated; better was intended

D. The tool refused

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A sales bot told "maximize sign-ups" signs people up without clear consent. What went wrong?

A. The goal lacked limits
B. The bot was too slow
C. People liked it
D. Nothing

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Which instruction better states intent?

A. "Make it shorter," which tells the tool exactly what the person wants in every way

B. "Fix it."

C. "Shorten this to 100 words, keeping all key facts."

D. "Do your best."

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A help-desk AI scored on "conversations ended quickly" starts ending chats early. What is this?

A. A bug in the keyboard
B. A legal requirement
C. Excellent service
D. A shortcut to the measure

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Which habit helps reduce alignment gaps with today's tools?

A. Testing for shortcuts

B. Never reviewing outcomes

C. Hiding the goal

D. Giving tools only one-word goals so that there is less for them to misunderstand

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Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PHIL.07 (local)

Learn
People disagree about how hard alignment will be for more capable systems and how much to worry, and this course does not settle that. Practical habits help with today's tools: state goals with limits and priorities, test for shortcuts, measure what you actually want, and keep humans reviewing important outcomes. The same gap appears with people and organizations, as when a measure becomes a target, so the idea is not unique to AI.

Do people agree on how hard alignment will be for more capable systems?

A. Yes, it is solved
B. No
C. It is never discussed
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-F2-Q1
Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PHIL.07 (local)

Learn
People disagree about how hard alignment will be for more capable systems and how much to worry, and this course does not settle that. Practical habits help with today's tools: state goals with limits and priorities, test for shortcuts, measure what you actually want, and keep humans reviewing important outcomes. The same gap appears with people and organizations, as when a measure becomes a target, so the idea is not unique to AI.

Does this course settle how much to worry about alignment?

A. It does not mention alignment
B. Yes, it says disaster is certain
C. No
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-F2-Q2
Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.PHIL.07 (local)

Learn
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Where else does the stated-versus-intended gap appear?

A. Only in math

B. Only in movies

C. Nowhere else, because only AI systems can ever miss the intent behind a stated goal

D. With people and organizations

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-F2-Q3
Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.PHIL.07 (local)

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Which practice helps today?

A. Stating goals with limits and priorities
B. Avoiding all goals
C. Never checking results
D. Stating goals in as few words as possible

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-F2-Q4
Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.PHIL.07 (local)

Learn
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Why keep humans reviewing important outcomes?

A. Because AI cannot produce outcomes
B. To catch shortcuts and missed intent
C. To slow things down
D. Because humans are always slower

REFERENCE LABEL: NOT PART OF THE QUESTION ID S146-F2-Q5
Artificial Intelligence, Set 146: Alignment as a concept · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.PHIL.07 (local)

Set 147: Weighing benefits and harms

Outcome AI.PHIL.08: Compare competing arguments about AI's role by their evidence and values, without treating a disputed view as fact

Module: Society: philosophy and social questions

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Debates about AI's role, for example in schools, art, medicine or work, often mix two things: EVIDENCE (what is true or likely) and VALUES (what matters most). Someone may favor AI tutors because studies show gains (evidence) and because they value access for all (value). Someone else may worry about lost skills (evidence question) and value human connection (value). To weigh arguments fairly, separate the two: check the evidence, name the values, and notice when a disputed view is presented as settled fact.

What two things do AI debates often mix?

A. Speed and size
B. Names and dates
C. Evidence and values
D. Prices and colors

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-M-Q1
Artificial Intelligence, Set 147: Weighing benefits and harms · MAIN · Q1/5 · Outcome AI.PHIL.08 (local)

Learn
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"Studies show test scores rose with AI tutors" is which kind of claim?

A. Evidence
B. A conclusion only
C. An opinion about art
D. A value

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-M-Q2
Artificial Intelligence, Set 147: Weighing benefits and harms · MAIN · Q2/5 · Outcome AI.PHIL.08 (local)

Learn
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"Human connection matters most in teaching" is which kind of claim?

A. A statistic
B. A law
C. Evidence
D. A value

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-M-Q3
Artificial Intelligence, Set 147: Weighing benefits and harms · MAIN · Q3/5 · Outcome AI.PHIL.08 (local)

Learn
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How should you weigh arguments fairly?

A. Ignore evidence

B. Check evidence, name values, and spot disputed views stated as fact

C. Ignore values

D. Decide which side has more people supporting it, and then accept that side's claims

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-M-Q4
Artificial Intelligence, Set 147: Weighing benefits and harms · MAIN · Q4/5 · Outcome AI.PHIL.08 (local)

Learn
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Which statement presents a disputed view as settled fact?

A. "Everyone knows AI art isn't real art."
B. "People disagree about AI art."
C. "A 2025 survey asked artists their views."
D. "Some artists argue AI art lacks intent."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-M-Q5
Artificial Intelligence, Set 147: Weighing benefits and harms · MAIN · Q5/5 · Outcome AI.PHIL.08 (local)

Learn
For each claim, ask: Could it be checked with data or observation? If yes, it is an EVIDENCE claim (true or false, strong or weak). If it is about what matters, what is good or what should be prioritized, it is a VALUE claim. Example debate about AI in hospitals: "AI reads scans faster" (evidence). "Speed matters more than cost" (value). "Patients deserve to know when AI is used" (value). "Most patients want to know" (evidence, checkable by surveys).

"AI reads scans faster" is which kind?

A. Neither
B. Both equally
C. A value
D. Evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-B1-Q1
Artificial Intelligence, Set 147: Weighing benefits and harms · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.PHIL.08 (local)

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"Speed matters more than cost" is which kind?

A. Neither
B. A statistic
C. Value
D. Evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-B1-Q2
Artificial Intelligence, Set 147: Weighing benefits and harms · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.PHIL.08 (local)

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"Patients deserve to know when AI is used" is which kind?

A. Neither
B. Value
C. A fact
D. Evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-B1-Q3
Artificial Intelligence, Set 147: Weighing benefits and harms · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.PHIL.08 (local)

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For each claim, ask: Could it be checked with data or observation? If yes, it is an EVIDENCE claim (true or false, strong or weak). If it is about what matters, what is good or what should be prioritized, it is a VALUE claim. Example debate about AI in hospitals: "AI reads scans faster" (evidence). "Speed matters more than cost" (value). "Patients deserve to know when AI is used" (value). "Most patients want to know" (evidence, checkable by surveys).

"Most patients want to know" is which kind?

A. Evidence
B. Neither
C. A law
D. A value

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-B1-Q4
Artificial Intelligence, Set 147: Weighing benefits and harms · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.PHIL.08 (local)

Learn
For each claim, ask: Could it be checked with data or observation? If yes, it is an EVIDENCE claim (true or false, strong or weak). If it is about what matters, what is good or what should be prioritized, it is a VALUE claim. Example debate about AI in hospitals: "AI reads scans faster" (evidence). "Speed matters more than cost" (value). "Patients deserve to know when AI is used" (value). "Most patients want to know" (evidence, checkable by surveys).

What question sorts the claims?

A. "Who said it?"
B. "Is it long?"
C. "Could it be checked with data or observation?"
D. "Do I personally agree with this claim?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-B1-Q5
Artificial Intelligence, Set 147: Weighing benefits and harms · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.PHIL.08 (local)

Learn

Debates about AI's role, for example in schools, art, medicine or work, often mix two things: EVIDENCE (what is true or likely) and VALUES (what matters most). Someone may favor AI tutors because studies show gains (evidence) and because they value access for all (value). Someone else may worry about lost skills (evidence question) and value human connection (value). To weigh arguments fairly, separate the two: check the evidence, name the values, and notice when a disputed view is presented as settled fact.

Two people agree AI tutors raise scores but disagree on using them. What is the likely source of disagreement?

A. A math error

B. A typo

C. Different evidence

D. Different values

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F1-Q1

Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.PHIL.08 (local)

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An article cites only studies that support its view. What is this?

A. Good research
B. Cherry-picking
C. A value claim
D. A balanced review

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F1-Q2
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.PHIL.08 (local)

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"AI will replace all teachers by 2030" is presented as fact. What is it really?

A. A law
B. A value
C. A prediction
D. A recorded fact

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F1-Q3
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.PHIL.08 (local)

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Someone says, "You only oppose AI because you are old." What is wrong?

A. It cites too much evidence
B. It dismisses the person instead of the argument
C. It is too polite
D. Nothing, because a person's age always tells you

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F1-Q4
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.PHIL.08 (local)

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Which summary is fairest?

A. The strongest version of each side

B. Only the other side

C. Only your side

D. The weakest version of the side you disagree with and the strongest version of your own side

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F1-Q5

Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.PHIL.08 (local)

Learn
Good-faith disagreement is normal when people share facts but weigh values differently. Evidence can be limited, early or mixed, so honest conclusions are often provisional. Watch for cherry-picking (only citing supportive studies), treating predictions as facts, and dismissing people instead of arguments. A fair summary states the strongest version of each side. This course does not tell you which side to take.

Evidence about a new AI use is early and mixed. What kind of conclusion is honest?

A. Whatever is popular

B. A final and certain conclusion, which should never be revisited even when new evidence appears

C. No conclusion ever

D. A provisional one

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F2-Q1
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.PHIL.08 (local)

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Is good-faith disagreement possible when people share the facts?

A. Only online

B. Only about sports

C. Yes, if they weigh values differently

D. No, people who share the same facts must always reach exactly the same conclusion

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F2-Q2
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.PHIL.08 (local)

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Which is a warning sign in an AI debate?

A. Checking sources

B. Naming values openly

C. Stating which studies support each side and noting where the evidence is still limited

D. Treating predictions as facts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F2-Q3
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.PHIL.08 (local)

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Does this course tell you which side to take on AI's role in society?

A. Yes, always the optimistic side
B. No
C. Yes, always the cautious side
D. Yes, it tells you exactly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F2-Q4
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.PHIL.08 (local)

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What does "strongest version of each side" require?

A. Understanding each side's best arguments
B. Ignoring evidence
C. Agreeing with everyone
D. Repeating the other side's worst arguments

REFERENCE LABEL: NOT PART OF THE QUESTION ID S147-F2-Q5
Artificial Intelligence, Set 147: Weighing benefits and harms · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.PHIL.08 (local)

Set 148: Early ideas and the name "AI"

Outcome AI.HISTORY.01: Place selected early ideas (Turing's 1950 paper, the 1955-56 Dartmouth project) and explain why they matter

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Two early moments shaped how people talk about AI. In 1950, Alan Turing's paper "Computing Machinery and Intelligence" asked whether machines could think and proposed the imitation game as a practical test. In August 1955, John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon wrote a proposal for a summer research project at Dartmouth College in 1956; the proposal is credited with introducing the term "ARTIFICIAL INTELLIGENCE." These moments matter because they framed AI as a research field and set questions people still debate.

What did Turing's 1950 paper propose?

A. The name "artificial intelligence"
B. A law about computers
C. The imitation game as a practical test
D. The first working chatbot

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-M-Q1
Artificial Intelligence, Set 148: Early ideas and the name "AI" · MAIN · Q1/5 · Outcome AI.HISTORY.01 (local)

Learn
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Which document is credited with introducing the term "artificial intelligence"?

A. A newspaper headline
B. A 2022 chatbot announcement
C. Turing's 1950 paper
D. The Dartmouth proposal

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-M-Q2
Artificial Intelligence, Set 148: Early ideas and the name "AI" · MAIN · Q2/5 · Outcome AI.HISTORY.01 (local)

Learn

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Prepared example (a short timeline from the course sources)

Year	Event
1950	Turing publishes "Computing Machinery and Intelligence"
1955	Dartmouth proposal written (dated August 31)
1956	Dartmouth summer research project held

In this timeline, how many years passed between Turing's paper and the Dartmouth project?

A. 6

B. 16

C. 1

D. 5, which is the number of years between Turing's paper and the date the proposal was written

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-M-Q3
Artificial Intelligence, Set 148: Early ideas and the name "AI" · MAIN · Q3/5 · Outcome AI.HISTORY.01 (local)

Learn
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Who was among the Dartmouth proposal's authors?

A. Garry Kasparov
B. John McCarthy
C. Joseph Weizenbaum
D. Alan Turing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-M-Q4
Artificial Intelligence, Set 148: Early ideas and the name "AI" · MAIN · Q4/5 · Outcome AI.HISTORY.01 (local)

Learn
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Why do these moments matter?

A. They framed AI as a field and set lasting questions
B. Because they ended AI research
C. Because they were secret
D. Because they produced the large language models

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-M-Q5
Artificial Intelligence, Set 148: Early ideas and the name "AI" · MAIN · Q5/5 · Outcome AI.HISTORY.01 (local)

Learn
To place events on a timeline, list them with their years, then sort them from earliest to latest. Compute gaps by subtracting. Example: Dartmouth project 1956; Turing's paper 1950; Dartmouth proposal 1955. Sorted: 1950, 1955, 1956. Gap from the paper to the proposal: 1955 - 1950 = 5 years. Gap from the proposal to the project: 1956 - 1955 = 1 year.

Which event is earliest?

A. All the same year

B. The proposal (1955)

C. The Dartmouth project (1956), since it is the most famous event in the whole timeline

D. Turing's paper (1950)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-B1-Q1
Artificial Intelligence, Set 148: Early ideas and the name "AI" · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.HISTORY.01 (local)

Learn
To place events on a timeline, list them with their years, then sort them from earliest to latest. Compute gaps by subtracting. Example: Dartmouth project 1956; Turing's paper 1950; Dartmouth proposal 1955. Sorted: 1950, 1955, 1956. Gap from the paper to the proposal: 1955 - 1950 = 5 years. Gap from the proposal to the project: 1956 - 1955 = 1 year.

Which event is latest?

A. The proposal (1955)

B. None

C. The Dartmouth project (1956)

D. Turing's paper (1950), because the oldest paper is always listed last on a timeline

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-B1-Q2
Artificial Intelligence, Set 148: Early ideas and the name "AI" · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.HISTORY.01 (local)

Learn
To place events on a timeline, list them with their years, then sort them from earliest to latest. Compute gaps by subtracting. Example: Dartmouth project 1956; Turing's paper 1950; Dartmouth proposal 1955. Sorted: 1950, 1955, 1956. Gap from the paper to the proposal: 1955 - 1950 = 5 years. Gap from the proposal to the project: 1956 - 1955 = 1 year.

Gap from the paper to the proposal?

A. 10 years
B. 5 years
C. 1 year
D. 6 years

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-B1-Q3
Artificial Intelligence, Set 148: Early ideas and the name "AI" · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.HISTORY.01 (local)

Learn
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Gap from the proposal to the project?

A. 6 years
B. 0 years
C. 1 year
D. 5 years

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-B1-Q4
Artificial Intelligence, Set 148: Early ideas and the name "AI" · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.HISTORY.01 (local)

Learn
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What is the sorted order?

A. 1950, 1955, 1956
B. 1955, 1950, 1956
C. 1950, 1956, 1955
D. 1956, 1955, 1950

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-B1-Q5
Artificial Intelligence, Set 148: Early ideas and the name "AI" · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.HISTORY.01 (local)

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A video claims "AI was invented in 2022." What is more accurate?

A. The video is correct, because nobody had any ideas about thinking machines before 2022

B. AI was invented in 1800

C. AI has no history

D. The term and field date to the 1950s

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F1-Q1
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.HISTORY.01 (local)

Learn
Two early moments shaped how people talk about AI. In 1950, Alan Turing's paper "Computing Machinery and Intelligence" asked whether machines could think and proposed the imitation game as a practical test. In August 1955, John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon wrote a proposal for a summer research project at Dartmouth College in 1956; the proposal is credited with introducing the term "ARTIFICIAL INTELLIGENCE." These moments matter because they framed AI as a research field and set questions people still debate.

Which question did Turing's paper start from?

A. "How can computers be made cheaper for schools and families?" which was a big concern then

B. "Can machines think?"

C. "How do phones work?"

D. "What is the internet?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F1-Q2
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.01 (local)

Learn
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Where was the 1956 summer project held?

A. Dartmouth College
B. A tech company
C. Online
D. MIT

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F1-Q3
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.HISTORY.01 (local)

Learn
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Why might the name "intelligence" shape expectations?

A. Because it is long

B. Because the name proves that every AI system thinks exactly the way people think

C. It invites comparisons with people

D. Because it is Latin

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F1-Q4
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.HISTORY.01 (local)

Learn
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Which is a lasting question from this era?

A. How to stream video

B. How to build apps

C. How many megabytes of memory a phone should have, which was the main debate in the 1950s

D. Whether machines can think

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F1-Q5
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.01 (local)

Learn
Early researchers were very optimistic: the Dartmouth proposal suggested significant progress could be made in one summer by a small group. Progress turned out to be much slower and uneven, with periods of reduced funding and interest. Remembering this helps when reading today's confident predictions. History also shows that a name can shape expectations: "intelligence" invites comparisons with people that the systems may not support.

How optimistic was the Dartmouth proposal?

A. It made no predictions

B. It suggested significant progress in one summer

C. It predicted that no progress at all would be made for at least one hundred years

D. It predicted 2022 chatbots

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F2-Q1
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.01 (local)

Learn
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How did progress actually go?

A. No progress ever

B. Instant success

C. Slower and uneven, with periods of reduced interest

D. Exactly as fast as the proposal predicted, with full human-like AI built by the late 1950s

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F2-Q2
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.01 (local)

Learn
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Why remember early optimism today?

A. Because it was secret

B. Because it is boring

C. Because it proves every prediction made about AI today will turn out to be wrong

D. It helps when reading confident predictions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F2-Q3
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.HISTORY.01 (local)

Learn
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Periods of reduced funding and interest in AI history are an example of what?

A. Uneven progress

B. A law

C. A product launch

D. Steady, uninterrupted progress that has continued at the same pace since the very beginning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F2-Q4
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.01 (local)

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Which lesson does this history support?

A. Ignore all research
B. Treat bold timelines with caution
C. History never repeats
D. Always believe the boldest timeline

REFERENCE LABEL: NOT PART OF THE QUESTION ID S148-F2-Q5
Artificial Intelligence, Set 148: Early ideas and the name "AI" · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.01 (local)

Set 149: From written rules to learning systems

Outcome AI.HISTORY.02: Explain the shift from mainly hand-written rules and search to learning from data, with dated examples

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Early AI often relied on HAND-WRITTEN RULES and SEARCH: people wrote rules, and programs searched through possibilities. IBM's Deep Blue beat world chess champion Garry Kasparov in a 1997 match, relying mainly on very fast search and evaluation methods crafted by experts. Over time, the field shifted toward LEARNING FROM DATA, where systems find patterns from examples. In March 2016, DeepMind's AlphaGo, which used deep neural networks trained on data and self-play, beat Go champion Lee Sedol 4 games to 1. Go has far more possible positions, so search alone was not enough.

What did early AI often rely on?

A. Large language models trained on most of the text on the internet, as modern chatbots are

B. Smartphones

C. Social media

D. Hand-written rules and search

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-M-Q1
Artificial Intelligence, Set 149: From written rules to learning systems · MAIN · Q1/5 · Outcome AI.HISTORY.02 (local)

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When did Deep Blue beat Garry Kasparov?

A. 1956
B. 1997
C. 2022
D. 2016

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-M-Q2
Artificial Intelligence, Set 149: From written rules to learning systems · MAIN · Q2/5 · Outcome AI.HISTORY.02 (local)

Learn
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What did Deep Blue mainly rely on?

A. Guessing
B. A chatbot
C. Very fast search and expert-crafted evaluation
D. Deep neural networks trained on millions of games

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-M-Q3
Artificial Intelligence, Set 149: From written rules to learning systems · MAIN · Q3/5 · Outcome AI.HISTORY.02 (local)

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What did AlphaGo use?

A. Deep neural networks trained on data and self-play, with search

B. A calculator

C. Nothing new

D. Only hand-written rules typed in by Go experts, with no learning from any data at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-M-Q4
Artificial Intelligence, Set 149: From written rules to learning systems · MAIN · Q4/5 · Outcome AI.HISTORY.02 (local)

Learn
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By what score did AlphaGo beat Lee Sedol in March 2016?

A. 1 game to 0
B. 3 games to 2
C. 5 games to 0
D. 4 games to 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-M-Q5
Artificial Intelligence, Set 149: From written rules to learning systems · MAIN · Q5/5 · Outcome AI.HISTORY.02 (local)

Learn
One cook follows a detailed RECIPE BOOK exactly: every step written by an expert. That works well for standard dishes but struggles with new ingredients. Another cook LEARNS BY TASTING thousands of dishes and noticing patterns, then improvises. That handles variety but may make surprising mistakes and cannot always explain why. Early AI was more like the recipe follower; much modern AI is more like the taster.

The recipe-book cook is like which approach?

A. A chatbot
B. Search only
C. Hand-written rules
D. Learning from data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-B1-Q1
Artificial Intelligence, Set 149: From written rules to learning systems · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.HISTORY.02 (local)

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The tasting cook is like which approach?

A. No approach
B. Learning from data
C. Search
D. Hand-written rules

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-B1-Q2
Artificial Intelligence, Set 149: From written rules to learning systems · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.HISTORY.02 (local)

Learn
One cook follows a detailed RECIPE BOOK exactly: every step written by an expert. That works well for standard dishes but struggles with new ingredients. Another cook LEARNS BY TASTING thousands of dishes and noticing patterns, then improvises. That handles variety but may make surprising mistakes and cannot always explain why. Early AI was more like the recipe follower; much modern AI is more like the taster.

Where does the recipe-book cook struggle?

A. With new ingredients
B. With reading
C. With ovens
D. With standard dishes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-B1-Q3
Artificial Intelligence, Set 149: From written rules to learning systems · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.HISTORY.02 (local)

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What is a weakness of the tasting cook?

A. Never being able to cook anything except the exact dishes written in a recipe book

B. Surprising mistakes and unclear reasons

C. Using too many pans

D. Being too slow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-B1-Q4
Artificial Intelligence, Set 149: From written rules to learning systems · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.HISTORY.02 (local)

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Much modern AI is more like which cook?

A. Both equally
B. Neither
C. The recipe-book cook
D. The tasting cook

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-B1-Q5
Artificial Intelligence, Set 149: From written rules to learning systems · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.HISTORY.02 (local)

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A tax calculator applies fixed rates by income. Which approach fits best?

A. Hand-written rules
B. Self-play
C. Random guessing
D. Learning from data

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F1-Q1
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.HISTORY.02 (local)

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A speech recognizer must handle many accents. Which approach fits best?

A. A calculator

B. No approach

C. Learning from data

D. Hand-written rules for every possible accent, written by experts one sound at a time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F1-Q2
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.02 (local)

Learn
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Why was search alone not enough for Go?

A. Because Go has no rules
B. Because computers cannot play games
C. Because Go is a very simple game
D. Go has far more possible positions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F1-Q3
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How many years passed between Deep Blue's win and AlphaGo's win?

A. 19

B. 16

C. 9, because Deep Blue won in the 1990s and AlphaGo won about ten years after that

D. 29

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F1-Q4
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.HISTORY.02 (local)

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Which statement about the shift is accurate?

A. Learning replaced everything
B. Rules replaced learning
C. It was gradual, and approaches are often combined
D. It was a sudden switch in one year, after

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F1-Q5
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.02 (local)

Learn
The shift was not a clean switch. Many systems combine rules, search and learning; AlphaGo itself combined learned networks with search. Rule-based systems are still useful where rules are clear and must be followed exactly, such as tax calculations. Learning systems shine where examples are plentiful but rules are hard to write, such as recognizing speech. Knowing which approach a system uses helps you predict how it will fail.

Did AlphaGo use only learning?

A. It used nothing
B. No; it combined learned networks with search
C. No, only rules
D. Yes, it used only neural networks and did not search

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F2-Q1
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.02 (local)

Learn
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Where are rule-based systems still useful?

A. Where rules are clear and must be followed exactly

B. Only in games

C. Nowhere, because rule-based systems were completely replaced by learning systems in 2016

D. Only in art

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F2-Q2
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.02 (local)

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Where do learning systems shine?

A. Only in chess

B. Where examples are plentiful but rules are hard to write

C. Where there are no examples at all and every rule is already written down clearly

D. Only in tax

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F2-Q3
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Why does knowing a system's approach help?

A. Because every approach fails in exactly the same way, so knowing it changes nothing

B. It changes the name

C. It sets the price

D. It helps predict how it will fail

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F2-Q4
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.02 (local)

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A learning system makes a strange mistake on an unusual input. Is this surprising given its approach?

A. Only rule systems err

B. Very surprising, since learning systems never make mistakes on any inputs whatsoever

C. Not very; learned patterns can fail on unusual cases

D. Impossible

REFERENCE LABEL: NOT PART OF THE QUESTION ID S149-F2-Q5
Artificial Intelligence, Set 149: From written rules to learning systems · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.02 (local)

Set 150: The deep learning and transformer era

Outcome AI.HISTORY.03: Describe selected milestones (2012 image recognition, the 2017 transformer, the 2022 chatbot release) and what they changed

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Three milestones show the deep learning era. In 2012, a deep neural network trained on graphics processors (GPUs), often called AlexNet, won a major image-recognition competition with a top-5 error of 15.3 percent versus 26.2 percent for the next entry. In 2017, the paper "Attention Is All You Need" introduced the TRANSFORMER, a design that became the basis of most later large language models. On November 30, 2022, OpenAI released ChatGPT as a free research preview, bringing chat-style access to a large language model to a very wide public.

What happened in 2012?

A. The first chatbot was released to the public and quickly gained millions of users worldwide

B. The transformer was introduced

C. A deep network won a major image-recognition competition

D. The term "AI" was coined

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-M-Q1
Artificial Intelligence, Set 150: The deep learning and transformer era · MAIN · Q1/5 · Outcome AI.HISTORY.03 (local)

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By how many percentage points did the 2012 winner beat the next entry's top-5 error?

A. 41.5
B. 26.2
C. 15.3
D. 10.9

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-M-Q2
Artificial Intelligence, Set 150: The deep learning and transformer era · MAIN · Q2/5 · Outcome AI.HISTORY.03 (local)

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What did the 2017 paper introduce?

A. GPUs
B. The transformer
C. The internet
D. The imitation game

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-M-Q3
Artificial Intelligence, Set 150: The deep learning and transformer era · MAIN · Q3/5 · Outcome AI.HISTORY.03 (local)

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When was ChatGPT released as a free research preview?

A. November 30, 2022

B. 1997

C. June 2017, which is when the transformer paper was first posted for researchers to read

D. 2012

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-M-Q4
Artificial Intelligence, Set 150: The deep learning and transformer era · MAIN · Q4/5 · Outcome AI.HISTORY.03 (local)

Learn

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Prepared example (milestones from the course sources)

Year	Milestone
2012	Deep network wins image-recognition competition
2017	Transformer paper published
2022	ChatGPT released as a free research preview

In this table, how many years separate the first and last milestones?

A. 5, which is the gap between the transformer paper and the chatbot release in the table

B. 10

C. 22

D. 12

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-M-Q5
Artificial Intelligence, Set 150: The deep learning and transformer era · MAIN · Q5/5 · Outcome AI.HISTORY.03 (local)

Learn
Cars changed in stages. A better ENGINE made cars powerful. A better DESIGN, such as the assembly line, made them practical to build at scale. Then affordable models put cars in ORDINARY FAMILIES' hands. AI's recent history has similar stages: 2012 showed deep networks were powerful, 2017 gave a design that scaled, and 2022 put chat-style AI in ordinary people's hands.

The 2012 result is most like which car stage?

A. A better engine

B. Road signs

C. The assembly line

D. Affordable cars for ordinary families, since 2012 was when everyone started using AI

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-B1-Q1
Artificial Intelligence, Set 150: The deep learning and transformer era · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.HISTORY.03 (local)

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The 2017 transformer is most like which stage?

A. Affordable cars
B. Gas stations
C. A better engine
D. A design that scaled

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-B1-Q2
Artificial Intelligence, Set 150: The deep learning and transformer era · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.HISTORY.03 (local)

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The 2022 release is most like which stage?

A. A better engine

B. The assembly line, since the release made it possible to build models at a large scale

C. Cars in ordinary families' hands

D. Traffic laws

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-B1-Q3
Artificial Intelligence, Set 150: The deep learning and transformer era · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.HISTORY.03 (local)

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Did each stage happen alone?

A. Yes, each stage happened suddenly without any connection to any earlier work at all

B. No; each built on earlier work

C. Yes, in one year

D. Only the last

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-B1-Q4
Artificial Intelligence, Set 150: The deep learning and transformer era · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.HISTORY.03 (local)

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What is the lesson?

A. Different milestones changed different things
B. Cars are AI
C. Only 2022 mattered
D. All three milestones changed exactly the same thing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-B1-Q5
Artificial Intelligence, Set 150: The deep learning and transformer era · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.HISTORY.03 (local)

Learn
Three milestones show the deep learning era. In 2012, a deep neural network trained on graphics processors (GPUs), often called AlexNet, won a major image-recognition competition with a top-5 error of 15.3 percent versus 26.2 percent for the next entry. In 2017, the paper "Attention Is All You Need" introduced the TRANSFORMER, a design that became the basis of most later large language models. On November 30, 2022, OpenAI released ChatGPT as a free research preview, bringing chat-style access to a large language model to a very wide public.

Someone says "ChatGPT invented the technology behind chatbots in 2022." What is more accurate?

A. That statement is completely accurate, since no earlier research was involved at all

B. Chatbots were invented in 2012

C. It built on earlier work such as the 2017 transformer

D. The transformer came in 2022

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F1-Q1
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.HISTORY.03 (local)

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What hardware helped the 2012 network?

A. Mobile phones, which were powerful enough in 2012 to train networks for image competitions

B. Calculators

C. Typewriters

D. Graphics processors (GPUs)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F1-Q2
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.03 (local)

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What did the 2022 release mainly change?

A. Who used the technology

B. Image recognition

C. Nothing

D. The basic design of language models, which was invented for the very first time that year

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F1-Q3
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Most later large language models are based on which design?

A. ELIZA
B. Deep Blue
C. The imitation game
D. The transformer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F1-Q4
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.HISTORY.03 (local)

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How many years after the transformer paper was ChatGPT released?

A. 17
B. 5
C. 2
D. 10

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F1-Q5
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.03 (local)

Learn
Each milestone changed something different. 2012 showed that deep networks plus lots of data and computing power could beat older methods on a hard task. 2017 gave a design that scaled well to huge amounts of text. 2022 changed who used the technology, putting it in everyday hands. None of these happened alone: they built on decades of earlier research, and each brought new questions about accuracy, cost and fairness.

What did 2012 show?

A. That GPUs were unnecessary
B. That chatbots were ready
C. Deep networks plus data and computing could beat older methods
D. That deep networks would never be useful for any task

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F2-Q1
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.03 (local)

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Why did the transformer matter for language models?

A. Because it was a chatbot

B. It scaled well to huge amounts of text

C. Because it was a law

D. Because it made models smaller and able to run on very old computers without any GPUs

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F2-Q2
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.03 (local)

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Did these milestones build on earlier research?

A. Yes, decades of it

B. Only the last one

C. Only the first one

D. No, each one appeared without any connection to anything that researchers had done before

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F2-Q3
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.HISTORY.03 (local)

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Which new questions did these milestones bring?

A. Only the color of computers

B. Only typing speed

C. Accuracy, cost and fairness

D. None at all, because each milestone solved every problem and raised no new questions

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F2-Q4
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.03 (local)

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Which statement fits this course's approach?

A. History does not matter
B. Dates are unknowable
C. Every important change in AI happened in a single year
D. Milestones are dated and each changed something specific

REFERENCE LABEL: NOT PART OF THE QUESTION ID S150-F2-Q5
Artificial Intelligence, Set 150: The deep learning and transformer era · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.03 (local)

Set 151: Narrow skill versus general intelligence claims

Outcome AI.HISTORY.04: Tell narrow task performance from claims of general intelligence

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A NARROW system does one task or a small set of tasks very well: Deep Blue played chess; AlphaGo played Go; an image classifier labels photos. GENERAL INTELLIGENCE, often called AGI (artificial general intelligence), would mean flexible ability across a wide range of tasks, including new ones, at or beyond human level; definitions vary and are debated. Strong performance on one task, or even many tests, does not by itself show general intelligence. Today's chat models handle many tasks, which makes the line blurrier, but they also fail in ways people do not expect.

What is a narrow system?

A. A slow system

B. A system that can do every task that any human can do, at least as well as any human

C. One that does one task or a small set very well

D. A small computer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-M-Q1
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · MAIN · Q1/5 · Outcome AI.HISTORY.04 (local)

Learn
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Which is an example of a narrow system?

A. A human chess player

B. The internet

C. A system that can learn any new task as flexibly as people can, without special training

D. Deep Blue

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-M-Q2
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · MAIN · Q2/5 · Outcome AI.HISTORY.04 (local)

Learn
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What would general intelligence (AGI) mean?

A. Being expensive
B. Flexible ability across many tasks, including new ones
C. Being fast
D. Being extremely good at exactly one game, such as chess

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-M-Q3
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · MAIN · Q3/5 · Outcome AI.HISTORY.04 (local)

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Does strong performance on one task show general intelligence?

A. No

B. Yes, if the task is hard

C. Yes, any system that is excellent at one task must be excellent at every other task

D. Only for games

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-M-Q4
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · MAIN · Q4/5 · Outcome AI.HISTORY.04 (local)

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Are definitions of AGI agreed?

A. Yes, there is one official definition of AGI that every researcher and company follows

B. There are none

C. No; they vary and are debated

D. Yes, by law

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-M-Q5
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · MAIN · Q5/5 · Outcome AI.HISTORY.04 (local)

Learn
A CALCULATOR multiplies huge numbers instantly, far better than any student. But it cannot read a story, learn a new game, or decide what problem to solve. A STUDENT is slower at multiplication but flexible: they can learn new subjects and handle new situations. Being superhuman at one task, like the calculator, is different from being flexible across many, like the student.

The calculator is like which kind of system?

A. Neither
B. Both
C. General
D. Narrow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-B1-Q1
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.HISTORY.04 (local)

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What can the calculator not do?

A. Learn a new game

B. Multiply large numbers quickly, which is the one thing that calculators struggle with

C. Turn on

D. Show numbers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-B1-Q2
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.HISTORY.04 (local)

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The student's strength is what?

A. Never learning
B. Flexibility across many tasks
C. Never making mistakes
D. Multiplying huge numbers faster

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-B1-Q3
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.HISTORY.04 (local)

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Is being superhuman at one task the same as being general?

A. Yes, for calculators
B. Only at night
C. No
D. Yes, any system that beats people at one task is general by definition

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-B1-Q4
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.HISTORY.04 (local)

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What is the lesson?

A. Narrow excellence is not general intelligence
B. Nothing
C. Students are calculators
D. Calculators are smarter than students at everything

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-B1-Q5
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.HISTORY.04 (local)

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A headline says "AI beats doctors at reading one type of scan, so AI is now generally intelligent." What is wrong?

A. Doctors cannot read scans

B. One task does not show general intelligence

C. Nothing, because beating doctors at one task proves the system can do any task at all

D. Scans are not tasks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F1-Q1
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A chat model handles many tasks but fails at simple new puzzles. What does this suggest?

A. That it cannot write
B. That puzzles are illegal
C. That it must be completely general
D. Its abilities are broad but uneven

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F1-Q2
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.04 (local)

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Which question best tests a general-intelligence claim?

A. "Is the logo new?"
B. "Were the tasks new to the system?"
C. "Is it expensive?"
D. "How many people liked the announcement?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F1-Q3
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Is AlphaGo an example of general intelligence?

A. Yes, it plays all games
B. It is a chatbot
C. No; it was built for Go
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F1-Q4
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Which statement fits this course's approach?

A. Experts disagree about how close current systems are to AGI

B. AGI is impossible, proven

C. Experts all agree that full AGI was achieved and announced to the public in early 2026

D. AGI exists already, proven

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F1-Q5
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.04 (local)

Learn
Claims about general intelligence deserve careful reading. Ask: How is "general" defined here? Which tasks were tested, and were they new to the system? How does it fail? Who is making the claim, and do they benefit from it? Experts disagree about how close current systems are to general intelligence and whether the concept is even well defined. This course does not settle that debate.

A company claims its product "reached AGI." What should you ask first?

A. "Can I buy it today at the lowest possible price, before everyone else hears about it?"

B. "What color is it?"

C. "Is it fast?"

D. "How is general defined here?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F2-Q1
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.04 (local)

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Why ask whether the claimant benefits?

A. Because companies always tell the exact truth about their own products in every announcement

B. Incentives can shape claims

C. Because it is polite

D. Because benefits are illegal

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F2-Q2
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.04 (local)

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Why ask how a system fails?

A. Because failures are rare

B. Because it is fun

C. Because systems that fail are always narrow, and systems that never fail are always general

D. Failures reveal limits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F2-Q3
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Do experts agree that AGI is well defined?

A. Yes, by law
B. Only in movies
C. No
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F2-Q4
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.04 (local)

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Does this course settle the AGI debate?

A. No
B. Yes, it says AGI exists
C. Yes, it says AGI is impossible
D. Yes, it states the exact year

REFERENCE LABEL: NOT PART OF THE QUESTION ID S151-F2-Q5
Artificial Intelligence, Set 151: Narrow skill versus general intelligence claims · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.04 (local)

Set 152: Forecasts, headlines and evidence

Outcome AI.HISTORY.05: Tell recorded events from predictions, and judge the evidence behind a capability claim or headline

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn

Headlines about AI mix RECORDED EVENTS ("a model was released on March 3"), CAPABILITY CLAIMS ("it solves difficult math problems") and PREDICTIONS ("AI will replace most jobs by 2030"). Each needs different checking. Events can be confirmed from records. Capability claims need EVIDENCE: what tests, how many, who ran them, compared with what? Predictions are uncertain by nature; check who is predicting, their track record, and what assumptions they make. A confident tone does not turn a prediction into a fact.

Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

In this fictional headline, which part is a recorded event?

A. Aces math olympiad problems

B. None

C. Will make math teachers obsolete

D. Released Tuesday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-M-Q1

Artificial Intelligence, Set 152: Forecasts, headlines and evidence · MAIN · Q1/5 · Outcome AI.HISTORY.05 (local)

Learn
Headlines about AI mix RECORDED EVENTS ("a model was released on March 3"), CAPABILITY CLAIMS ("it solves difficult math problems") and PREDICTIONS ("AI will replace most jobs by 2030"). Each needs different checking. Events can be confirmed from records. Capability claims need EVIDENCE: what tests, how many, who ran them, compared with what? Predictions are uncertain by nature; check who is predicting, their track record, and what assumptions they make. A confident tone does not turn a prediction into a fact.

Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

In the same headline, which part is a capability claim?

A. None

B. Released Tuesday, which describes what the new AI system is able to do for its users

C. Aces math olympiad problems

D. Obsolete by 2030

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-M-Q2
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · MAIN · Q2/5 · Outcome AI.HISTORY.05 (local)

Learn
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Prepared example (a fictional headline)

Headline: "New AI released Tuesday aces math olympiad problems; experts say it will make math teachers obsolete by 2030."

In the same headline, which part is a prediction?

A. Released Tuesday, since it describes what will happen to the system in the near future

B. Teachers obsolete by 2030

C. Aces math problems

D. None

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-M-Q3
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · MAIN · Q3/5 · Outcome AI.HISTORY.05 (local)

Learn
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What do capability claims need?

A. Evidence: which tests, how many, who ran them

B. A famous logo

C. A long article

D. A confident tone and an exciting headline

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-M-Q4
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · MAIN · Q4/5 · Outcome AI.HISTORY.05 (local)

Learn
Headlines about AI mix RECORDED EVENTS ("a model was released on March 3"), CAPABILITY CLAIMS ("it solves difficult math problems") and PREDICTIONS ("AI will replace most jobs by 2030"). Each needs different checking. Events can be confirmed from records. Capability claims need EVIDENCE: what tests, how many, who ran them, compared with what? Predictions are uncertain by nature; check who is predicting, their track record, and what assumptions they make. A confident tone does not turn a prediction into a fact.

Does a confident tone turn a prediction into a fact?

A. Only in headlines
B. Yes, if repeated
C. No
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-M-Q5
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · MAIN · Q5/5 · Outcome AI.HISTORY.05 (local)

Learn
Sort each statement. RECORDED EVENT: something that happened, checkable in records ("signed on May 1"). CAPABILITY CLAIM: what a system can do, which needs test evidence ("translates 100 languages"). PREDICTION: what will happen ("will cure diseases within five years"). Example: "Launched last week, the app detects skin cancer as well as doctors and will save millions of lives." Event: launched last week. Capability: detects as well as doctors. Prediction: will save millions.

"Launched last week" is which kind?

A. Value
B. Capability claim
C. Prediction
D. Recorded event

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-B1-Q1
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.HISTORY.05 (local)

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"Detects skin cancer as well as doctors" is which kind?

A. Capability claim
B. Prediction
C. Value
D. Recorded event

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-B1-Q2
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.HISTORY.05 (local)

Learn
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"Will save millions of lives" is which kind?

A. Recorded event
B. Prediction
C. Value
D. Capability claim

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-B1-Q3
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.HISTORY.05 (local)

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What checks a recorded event?

A. Records

B. Tone

C. Likes

D. Expert predictions about the future, which confirm whether an event really happened

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-B1-Q4
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.HISTORY.05 (local)

Learn
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What checks a capability claim?

A. The number of times the claim has been repeated in different news articles online

B. Predictions

C. Records of the launch date

D. Test evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-B1-Q5
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.HISTORY.05 (local)

Learn
Headlines about AI mix RECORDED EVENTS ("a model was released on March 3"), CAPABILITY CLAIMS ("it solves difficult math problems") and PREDICTIONS ("AI will replace most jobs by 2030"). Each needs different checking. Events can be confirmed from records. Capability claims need EVIDENCE: what tests, how many, who ran them, compared with what? Predictions are uncertain by nature; check who is predicting, their track record, and what assumptions they make. A confident tone does not turn a prediction into a fact.

A company says its model "matches experts," based on its own unpublished test. How strong is this evidence?

A. Irrelevant

B. Weak until details or independent checks appear

C. Very strong, because companies always test their own products in the most careful possible way

D. Proof

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F1-Q1
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.HISTORY.05 (local)

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A prediction says "AI will do X by 2030." What should you check?

A. The font
B. Nothing
C. Who predicts, their track record and assumptions
D. Whether the prediction has been shared widely

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F1-Q2
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.05 (local)

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A demo video shows a robot folding laundry perfectly. What should you consider?

A. Nothing

B. It may show the best attempt out of many

C. That robots cannot fold

D. That the robot folds every piece of laundry perfectly every single time in any home

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F1-Q3
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.HISTORY.05 (local)

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Other researchers repeat a test and get the same result. What does this do?

A. Nothing
B. Proves it forever
C. Weakens the claim
D. Strengthens the claim

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F1-Q4
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.HISTORY.05 (local)

Learn
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What is the first step with a surprising AI headline?

A. Ignore all news
B. Believe it
C. Find the original source
D. Share it with friends quickly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F1-Q5
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.05 (local)

Learn
History offers perspective: confident predictions about AI have been wrong in both directions, too early and too late. Demos may show the best attempt out of many. Press releases may omit failures. Independent checks, such as other researchers repeating a test, strengthen a claim. When reading a headline, find the original source, separate the three kinds of statements, and give each the trust its evidence earns.

Have AI predictions historically been wrong?

A. Yes, in both directions

B. Only too early

C. Only too late

D. No, every prediction about AI has come true exactly on the date that was predicted

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F2-Q1
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.05 (local)

Learn
History offers perspective: confident predictions about AI have been wrong in both directions, too early and too late. Demos may show the best attempt out of many. Press releases may omit failures. Independent checks, such as other researchers repeating a test, strengthen a claim. When reading a headline, find the original source, separate the three kinds of statements, and give each the trust its evidence earns.

What might a press release leave out?

A. The company name
B. Failures
C. The date
D. Every success

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F2-Q2
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.05 (local)

Learn
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How should trust be assigned to statements in a headline?

A. None at all
B. Fully to predictions
C. Equally to every statement
D. Each according to its evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F2-Q3
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.HISTORY.05 (local)

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Which strengthens a capability claim most?

A. The company's chief executive repeating the claim confidently in several interviews

B. More advertising

C. Independent researchers repeating the test

D. A larger font

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F2-Q4
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.05 (local)

Learn
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Which lesson does AI history support about predictions?

A. Read them with caution

B. Believe the most confident ones, since confidence is the best sign of an accurate prediction

C. Never read them

D. They are always right

REFERENCE LABEL: NOT PART OF THE QUESTION ID S152-F2-Q5
Artificial Intelligence, Set 152: Forecasts, headlines and evidence · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.05 (local)

Set 153: Energy and resource use

Outcome AI.HISTORY.06: Describe energy and resource demands of AI using dated, sourced estimates and their uncertainty, without one universal per-question number

Module: Society: history and claims about the future

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
AI uses ENERGY and other RESOURCES: electricity to train and run models in DATA CENTRES (buildings full of computers), water for cooling in some designs, and materials for chips. The International Energy Agency (IEA) estimated that data centres used about 415 terawatt-hours (TWh) of electricity in 2024, about 1.5 percent of global use, and projected around 945 TWh by 2030 in its base case, stressing large uncertainty. Data centres serve much more than AI. Per-question energy figures vary widely and should not be treated as one universal number.

What is a data centre?

A. A building full of computers

B. A school computer lab

C. A website

D. A shopping center where people buy computers and phones for their homes and offices

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-M-Q1
Artificial Intelligence, Set 153: Energy and resource use · MAIN · Q1/5 · Outcome AI.HISTORY.06 (local)

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Per the IEA, about how much electricity did data centres use in 2024?

A. Exactly 0 TWh
B. About 1.5 TWh
C. About 945 TWh
D. About 415 TWh

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-M-Q2
Artificial Intelligence, Set 153: Energy and resource use · MAIN · Q2/5 · Outcome AI.HISTORY.06 (local)

Learn
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About what share of global electricity was that?

A. About 15 percent
B. About 0.01 percent
C. About 1.5 percent
D. About 50 percent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-M-Q3
Artificial Intelligence, Set 153: Energy and resource use · MAIN · Q3/5 · Outcome AI.HISTORY.06 (local)

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From 415 to 945 TWh is roughly what increase?

A. About 2.3 TWh
B. About 530 TWh
C. About 945 TWh
D. About 1,360 TWh

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-M-Q4
Artificial Intelligence, Set 153: Energy and resource use · MAIN · Q4/5 · Outcome AI.HISTORY.06 (local)

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Why not quote one universal per-question energy number?

A. Because AI uses no energy

B. Because energy is not measured

C. Because every AI question uses exactly the same energy, so no number is needed at all

D. Figures vary widely

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-M-Q5
Artificial Intelligence, Set 153: Energy and resource use · MAIN · Q5/5 · Outcome AI.HISTORY.06 (local)

Learn
How much fuel does a car trip use? It depends on the car, the distance, the speed, traffic and the weather. A single number like "a trip uses 5 liters" is meaningless without those details. AI energy is similar: the energy for a question depends on the model, the hardware, the data centre and the power source. Good estimates state their assumptions and ranges.

What does a trip's fuel use depend on?

A. Car, distance, speed, traffic and weather
B. Only the driver's name
C. Nothing
D. Only the color of the car, which always decides how much fuel any trip will use

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-B1-Q1
Artificial Intelligence, Set 153: Energy and resource use · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.HISTORY.06 (local)

Learn
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Is "a trip uses 5 liters" meaningful alone?

A. Yes, for short trips
B. No; details are missing
C. Only in summer
D. Yes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-B1-Q2
Artificial Intelligence, Set 153: Energy and resource use · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.HISTORY.06 (local)

Learn
How much fuel does a car trip use? It depends on the car, the distance, the speed, traffic and the weather. A single number like "a trip uses 5 liters" is meaningless without those details. AI energy is similar: the energy for a question depends on the model, the hardware, the data centre and the power source. Good estimates state their assumptions and ranges.

In AI, what is like the car?

A. The person asking the question, whose typing speed decides the energy that is used

B. The question's topic

C. The model and hardware

D. The weather

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-B1-Q3
Artificial Intelligence, Set 153: Energy and resource use · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.HISTORY.06 (local)

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What should good estimates state?

A. Only the source's logo
B. Only the date
C. Only one exact number
D. Assumptions and ranges

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-B1-Q4
Artificial Intelligence, Set 153: Energy and resource use · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.HISTORY.06 (local)

Learn
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What is the lesson?

A. Energy figures depend on many factors

B. Cars use no fuel

C. AI uses no energy

D. AI energy use can be described by one exact universal number for every question

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-B1-Q5
Artificial Intelligence, Set 153: Energy and resource use · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.HISTORY.06 (local)

Learn
AI uses ENERGY and other RESOURCES: electricity to train and run models in DATA CENTRES (buildings full of computers), water for cooling in some designs, and materials for chips. The International Energy Agency (IEA) estimated that data centres used about 415 terawatt-hours (TWh) of electricity in 2024, about 1.5 percent of global use, and projected around 945 TWh by 2030 in its base case, stressing large uncertainty. Data centres serve much more than AI. Per-question energy figures vary widely and should not be treated as one universal number.

A post says "Every AI question uses exactly one bottle of water." What should you check?

A. The bottle's brand

B. Its source, date and assumptions

C. Nothing, since a claim that is easy to remember must be accurate for every AI question

D. The post's likes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F1-Q1
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.HISTORY.06 (local)

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Do data centres serve only AI?

A. Yes, every data centre in the world was built only to run AI chatbots and nothing else

B. Only in Europe

C. No; they serve much more

D. Yes, since 2022

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F1-Q2
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.HISTORY.06 (local)

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The IEA's 2030 figure is what kind of statement?

A. A law

B. A projection with large uncertainty

C. A recorded fact about electricity use that has already been measured in the year 2030

D. A guess with no method

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F1-Q3
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A student writes "AI uses 1.5% of world electricity." What is the correction?

A. The IEA estimate is for all data centres, not just AI

B. AI uses none

C. No correction is needed, since every data centre is used only for AI and nothing else

D. It is 50%

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F1-Q4
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Which resource besides electricity can AI use?

A. Gasoline, which every data centre burns directly to power the computers that run AI

B. Sand only

C. Water for cooling in some designs

D. Paper

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F1-Q5
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.HISTORY.06 (local)

Learn
Estimates differ because they depend on the model size, hardware, how busy data centres are, the electricity source, and what is counted, such as training versus everyday use. Newer hardware can use less energy per task, while growing use can raise the total. Claims like "each question uses a bottle of water" may be based on particular assumptions. Look for the source, date, what was measured and the range of uncertainty.

Which factor affects AI energy estimates?

A. The user's age
B. The day of the week only
C. The number of letters
D. The electricity source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F2-Q1
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.HISTORY.06 (local)

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How can newer hardware affect energy?

A. It has no effect

B. It can use less energy per task

C. It uses no energy

D. It always uses far more energy per task, since newer computers are always less efficient

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F2-Q2
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.HISTORY.06 (local)

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Why might total energy still rise with more efficient hardware?

A. It cannot rise

B. Because efficient hardware always uses more energy in total than older, less efficient hardware

C. Growing use can raise the total

D. Because energy is free

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F2-Q3
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.HISTORY.06 (local)

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Training versus everyday use: why does this matter for estimates?

A. Because training is free

B. It does not matter

C. Because training and everyday use always consume exactly the same amount of energy

D. They count different things

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F2-Q4
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.HISTORY.06 (local)

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What should you look for in an energy claim?

A. Source, date, what was measured and uncertainty
B. Only a single number
C. Only the headline
D. Only how alarming or reassuring the claim sounds

REFERENCE LABEL: NOT PART OF THE QUESTION ID S153-F2-Q5
Artificial Intelligence, Set 153: Energy and resource use · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.HISTORY.06 (local)